

Solid-State Drive: Ubiquitous Device in Making

With the tsunami of data in every application, a highly efficient data storage drive is going to be the de facto choice for all future gadgets, starting from mobiles, laptops, and mini/microservers to high-performance computing systems.



KIOXIA Demonstrates New EDSFF SSD Form Factor Purpose-Built for Servers and Storage



The various form factors of SSD drives

Non-volatile memory express (NVMe)-based solid-state drive (SSD) is a feature-rich device that can be exploited for future scaling needs via a peripheral component interconnect express (PCIe) bus for maximum throughput and lower latency. The recent trend of a further reduction in latency for hyperconvergence of servers in high-performance computing (HPC) systems increases its potential to be the ubiquitous candidate of the future.

SSD and its inherent advantages

Before going into the details of what factors are in favour of SSD devices to be the most probable candidate of the future, we should first take a look at the importance of SSD as a storage device.

An SSD's faster boot speeds and orders of magnitude increase in the read/write speeds benefit an individual or an enterprise in saving time and improving productivity. The newer applications like streaming 4K/8K videos and the use case of 5G/6G communications will find SSD a better match to address the bottlenecks as a storage partner in the essential chain

of computer+network+storage. A data-centre-ready/cloud-ready infrastructure can help transform the economics for maintaining such a converged infrastructure. The following are three major components in revolutionising the future server farms.

Computing server farms: Over the years, we have experienced the explosion of data that may be further fuelled by technological revolutions such as the Internet of Things (IoT), artificial intelligence (AI), blockchain, autonomous driving, 4K/8K video streaming, 5G networks, etc. This has led to significant advancements in computing. To further reduce the latency in the cluster of computing cores, PCIe controller, network-on-chip (NOC), and the new standards like Compute Express Link (CXL) are being used. With the data-intensive advanced applications like AI and camera interface with HD videos, the load on the Internet has been increasing.

Of late, edge computing has become a prominent accelerator for AI processing and ML by reducing the amount of data transfer on the net. This has been possible by in-memory

computing or analysing the data through in-built analysers attached to the storage devices and computational storage drives (CSD).

Networking: The demand for the network bandwidth continues to grow exponentially with emerging applications like AI, high-bandwidth 5G use cases, ML, IoT, and edge computing applications. Huge demand for throughput with low latency is being addressed with various Physical Media Architectures of SerDes (Serialiser and Deserialiser), which is a cornerstone block for high-speed serial links.

The data can be transferred over optical/electrical cables or short PCB tracks. Various standard bodies are working on these SerDes specifications to support the ever-increasing demand for higher bandwidth and lower latency of the data centres, cloud and edge computing, 5G, AI, and ML applications.

Storage: The evolution of storage devices over the past seven decades witnessed continued miniaturisation of the storage drives, from a wardrobe sized drive that contains a few megabytes of data to a single-chip device that contains a few terabytes of data. The advancements